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Dr. Michael Rowan
Executive Editor, Green Chemistry

Manuscript submission

Dear Dr. Rowan, dear Michael,

I am pleased to submit the paper **Analysis of nitrous oxide as a green oxidant based on economic, environmental, and safety metrics** by A. Nabera, G.-M. Beshara, G. Guillén-Gosálbez, and J. Pérez-Ramírez, for consideration in Green Chemistry.

The pressing demand for sustainable chemical synthesis calls for atom-efficient and selective oxidation strategies. Nitrous oxide, N_2O, is an attractive candidate, capable of delivering mono-oxygen species with high selectivity. Industrially, however, its production relies on the five-step thermal decomposition of ammonium nitrate, a route burdened by safety, environmental, and economic drawbacks. Recent progress in catalyst design highlights the promise of a one-step ammonia oxidation pathway as a more streamlined and cost-effective alternative. Yet, systematic metrics for benchmarking N_2O production technologies against established oxidants such as hydrogen peroxide (H_2O_2) remain underdeveloped, rendering the one-step route's benefits largely speculative. Moreover, most existing evaluation frameworks neglect to integrate the three fundamental pillars of sustainability—economic viability, environmental performance, and safety considerations of societal impact.

This study introduces a framework to evaluate emerging technologies across all three sustainability pillars by integrating techno-economic analysis, life cycle assessment, and quantitative safety indicators, including Dow's Fire & Explosion Index and TNT equivalency. Focusing on N_2O, we first compare the one-step process to the conventional five-step route, showing that the one-step route—using fossil natural gas or green, electrolytic hydrogen-based ammonia—reduces costs and carbon footprint by over 20%, while significantly lowering safety hazard potential.

We further compare N_2O produced via the one-step process with H_2O_2, a well-established oxidant produced via the anthraquinone process. Using our integrated framework, we show that N_2O from a fossil/green ammonia blend achieves a similar carbon footprint while offering ~40% cost savings and improved safety.

Lastly, given the benefits of one-step N_2O, we evaluate a key application, i.e., phenol production via the N_2O-mediated direct oxidation of benzene, and compare it to the traditional cumene oxidation process as well as the direct oxidation of benzene using H_2O_2. This case study highlights that N_2O-based direct oxidation enables a cost-effective and lower carbon footprint route compared to the cumene process, while economically outperforming H_2O_2-based direct oxidation of benzene to produce phenol.

Overall, these findings reinforce the potential of N_2O in oxidation chemistry and highlight the importance of our multidimensional, integrated sustainability framework as a template for evaluating novel technologies. Consequently, we are confident that this work will strongly appeal to the readership of Green Chemistry. Please find attached the manuscript files, and we look forward to your kind consideration.

Best wishes,

Javier Pérez-Ramírez